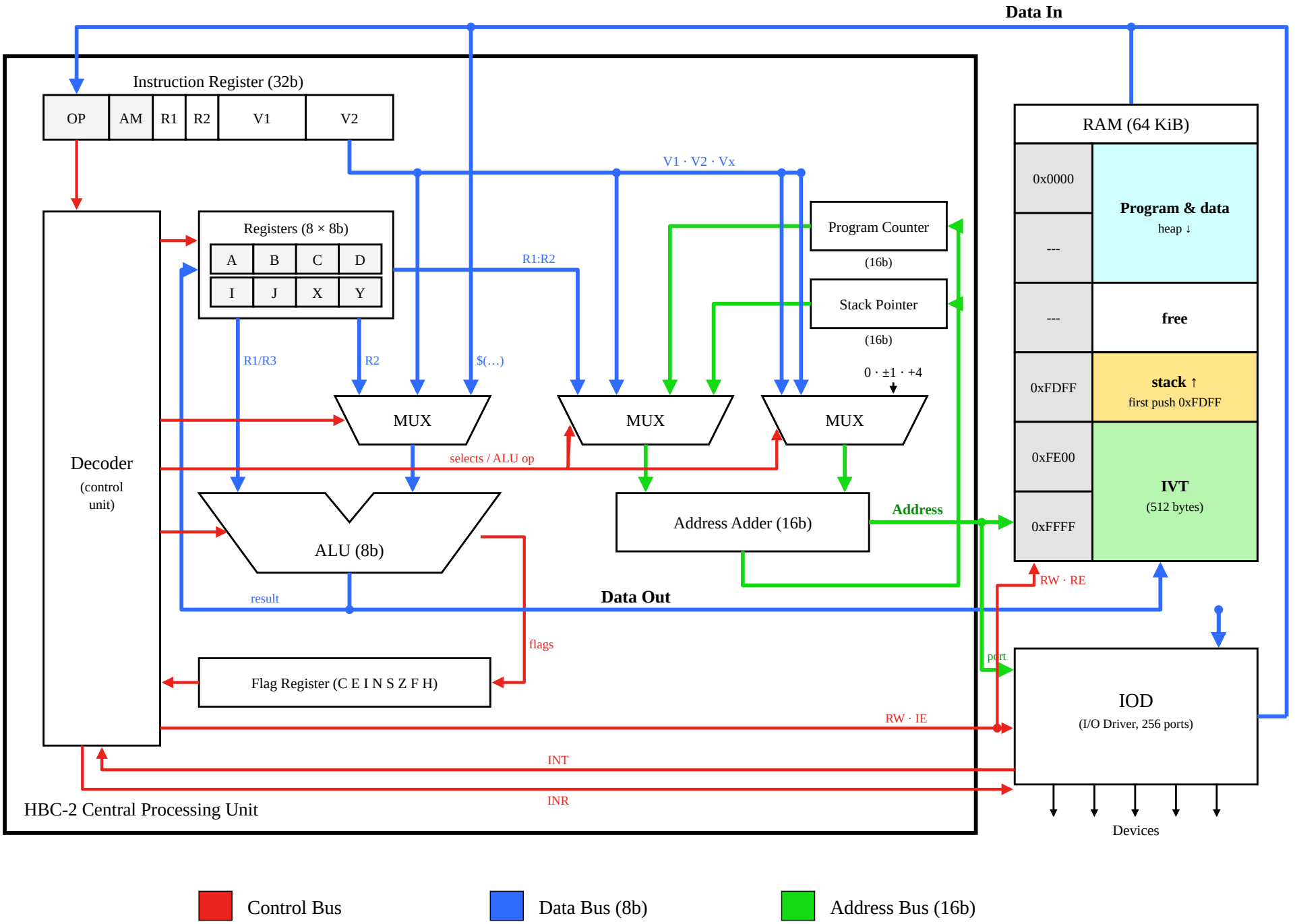
Data In
Instruction Register (32b)
OP
AM
R1
R2
V1
V2
V1 · V2 · Vx
Registers (8 × 8b)
A
B
C
D
I
J
X
Y
R1:R2
Program Counter
(16b)
Stack Pointer
(16b)
R1/R3
R2
$(…)
0 · ±1 · +4
MUX
MUX
MUX
Decoder
(control unit)
selects / ALU op
ALU (8b)
Address Adder (16b)
Address
result
Data Out
flags
Flag Register (C E I N S Z F H)
RW · RE
port
RW · IE
INT
INR
HBC-2 Central Processing Unit
RAM (64 KiB)
0x0000
Program & data
heap ↓

---
free
0xFDFF
stack ↑
first push 0xFDFF
0xFE00
IVT
(512 bytes)
0xFFFF
IOD
(I/O Driver, 256 ports)
Devices
Control Bus
Data Bus (8b)
Address Bus (16b)